

CAPSTONE 2

PROGRESS REPORT 1

ColdTrace: AI-Assisted Cold-Chain Monitoring and Delivery Management System

Completed PHP/Laravel modules verified from the current project repository

Project Information	Details
Project Title	ColdTrace: AI-Assisted Cold-Chain Monitoring and Delivery Management System
Project Leader	[Insert project leader name]
Group Members	[Insert group member names]
Technical Adviser	[Insert technical adviser name]
Reporting Date	August 15, 2026

REPORT SCOPE

This report lists only functions that are implemented in the current ColdTrace codebase. Incomplete or placeholder areas are intentionally omitted, and the listed scope is not an overall project-completion percentage.

1. Completed System Functionality / Features / Modules

The register below contains only modules and deliverables that are present in the current implementation. Each listed item is marked completed within this report's defined scope.

No.	Completed Module	Implemented Functionality	Status
1	Authentication and Role Access	Active-account login, secure logout and session invalidation, role-based redirects, and server-side access checks for Administrator, Driver, and Receiver accounts.	COMPLETED
2	Administrator Dashboard and Fleet Overview	Operational counts for users, active devices, trips, and unresolved critical alerts, together with fleet and order monitoring based on the latest available telemetry.	COMPLETED
3	User Account Management	Administrator-only account create, view, edit, and delete functions; role and status assignment; search and filters; summary counts; and protection against deleting the signed-in account.	COMPLETED
4	Administrator Order Management and Assignment	Create, view, edit, cancel, and delete orders with multiple products, receiver and active-driver assignment, schedule conflict validation, trip synchronization, and driver assignment notification.	COMPLETED
5	Receiver Order Portal	Receiver dashboard, order search and status totals, multi-item order creation, ownership checks, pending-order editing or deletion, permitted cancellation, and saved delivery address and map location.	COMPLETED
6	Driver Dashboard, Orders, and Notifications	Driver-specific trip totals, assigned-order search and filters, current assignment details, unread assignment notifications, mark-one or mark-all read actions, and order-specific latest telemetry access.	COMPLETED
7	Delivery Route Planning and Map Visualization	Mapped delivery destinations, multi-stop route construction, route alternatives, ETA and distance display, optimized stop ordering, and browser location fallback when vehicle telemetry is unavailable.	COMPLETED
8	Six-Device ESP32 Registry and MQTT Topics	Server configuration and database seeding for ESP32-CT-1001 through ESP32-CT-1006, matching per-device telemetry topics, optional truck assignment, and a shared wildcard subscription topic.	COMPLETED
9	Telemetry Intake and Device Heartbeat	Validated POST /api/telemetry intake for configured devices; persistence of temperature, humidity, GPS, signal, and battery readings; device last-seen and active-status updates; and active-trip association.	COMPLETED
10	Cold-Chain Analytics	Mean Kinetic Temperature is calculated from recorded trip temperatures, Remaining Shelf Life is estimated for the assigned product, and both values are stored with telemetry for monitoring use.	COMPLETED
11	Server-Authoritative Route Risk and Scoring	PHP reloads the authenticated driver's order, trip, product, device, and latest database telemetry; verifies ownership and device-truck linkage; then computes temperature risk, RSL risk, and a normalized route score.	COMPLETED
12	Validated AI Route Recommendation and Fallback	The AI may select only a submitted route ID. Response structure, risk level, and confidence are validated, confidence is clamped to 0-1, and API, timeout, connection, or malformed-response failures return the deterministic lowest server score.	COMPLETED

No.	Completed Module	Implemented Functionality	Status
13	Administrative Reports and CSV Export	Date-range statistics, order and trip status summaries, driver performance, daily temperature trends, temperature-breach counts, and CSV export for orders, trips, alerts, and telemetry.	COMPLETED
14	Automated Verification	The current suite passes 11 tests with 49 assertions, covering the six-device configuration, telemetry route and fleet rejection, server-owned route scoring, cross-truck telemetry rejection, AI response validation and fallback, and guest login redirection.	COMPLETED

2. Verification Summary

Verification was performed using the project's existing automated test suite on August 15, 2026. The suite completed successfully with the following result:

Verification Item	Verified Result
Test result	PASS
Tests	11 passed
Assertions	49 passed
Device profiles verified	ESP32-CT-1001 through ESP32-CT-1006
Key safeguards verified	Server-owned route scores, device-truck ownership, AI route validation, and deterministic fallback

3. Review and Sign-off

The signatures below confirm review of the completed-module register for this reporting period.

Prepared by	Checked by
[Project Leader / Group Representative]	[Capstone 2 Subject Teacher]
Signature: _____	Signature: _____
Date: _____	Date: _____